

Sakib Mohammed Sobaha

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Personal Profile

A Computer Science and Engineering undergraduate with a strong passion for problem-solving, algorithms, and mathematics. Enthusiastic about machine learning, cybersecurity, and software development, with a keen interest in Data Structures & Algorithms (DSA). Dedicated to continuous learning and knowledge sharing.

Experience

Synesis IT PLC

Software Engineer, AI and R&D Team

Apr 2025 – Present

- Developing multilingual Retrieval-Augmented Generation (RAG) chatbots with FastAPI, Hugging Face Transformers, and FAISS.
- Experimented with open-source LLMs including Mistral, Qwen, and Gemma to benchmark and improve chatbot response accuracy and relevance.
- Integrated reranking mechanisms and fine-tuned a BanglaBERT model for domain-specific text classification.
- Worked on GraphRAG integration for enhancing retrieval by auto-generating query variations.
- Worked on preprocessing Election Commission, Land, and Job Portal datasets for question answering and information retrieval.
- Contributed to deployment, optimization, and latency reduction of ASR-RAG pipelines in production environments.
- Implemented semantic query understanding with embeddings and cross-encoders to retrieve predefined answers from domain-specific datasets.

Education

Bangladesh University of Engineering and Technology

3.65/4.00

BSc in Computer Science and Engineering

Feb 2020 - Mar 2025

- Notable Courses:** Data Structure and Algorithm, Computer Security, Software Development, Basic Graph Theory, Machine Learning, Linear Algebra, Computer Architecture

Notre Dame College

5.00/5.00

HSC (Science)

2017 - 2019

- 22nd among 2027 students
- General Grade Scholarship

Ideal School and College

5.00/5.00

SSC (Science)

2007 - 2017

- Talentpool Scholarship in PSC, JSC & SSC
- 5th in Dhaka Board in SSC

Research Experience

University of Illinois Urbana-Champaign

Remote

Summer Research Intern

Jun. 2026 – Present

- Selected for the UIUC++ SRSE 2026 program.
- Actively contributing to research initiatives centered on large language models (LLMs), including literature review, experimentation, and result analysis.

Bangla Fake News Detection

Undergraduate Thesis

- Development of a Bangla Fake News Detection Framework.
- Created a dedicated dataset for misinformation and disinformation detection.
- Comparative Analysis of Machine Learning and Deep Learning Models.
- Enhanced Preprocessing Techniques for Bangla Text.

Publications

CONFERENCE PROCEEDINGS

- Sakib Mohammed Sobaha**, Nazmus Sakib Sami, Sadia Sharmin. *Bangla Fake News Detection with Source-Specific Probabilities for Improved Accuracy*. In *HCI International 2025 Posters*, Washington, D.C., USA. Published by Springer, Cham, 2025. eBook ISBN: 978-3-031-94153-5, Print ISBN: 978-3-031-94152-8. DOI [DOI](#)

Achievements

2025	2nd Runners Up , UIU Project ShowCase Competition	UIU CSE FEST
2024	Posters on Bangla Fake News Detection got accepted , Nsyss 2024	BUET Nsyss 2024
2024	2nd Runners Up , Therap Java Fest	Therap BD Ltd
2021	Champion & Speaker of the final , BUETDC Freshers' 2021	BUET Debating Club
2021	Champion , BUET EDC Case Competition	
2020	University Dean's List , Level 1 Term 2	
2018	7th in Dhaka Division , Bangaldesh Chemistry Olympiad	BChO 2018 Dhaka

Projects

Pet Palok, A Pet Community Web App

Software Project | [Project Link](#) | [Visit PetPalok](#)

- **Pet Profiles & Services:** Manage pet profiles, gallery, medical history, and schedule vet appointments (online via PetPalok).
- **Marketplace:** Platform for sellers and users to trade pet goods, with price prediction using TensorFlow GRU.
- **Community & Chat:** Facebook-like groups for updates, user-to-user chat (Socket Programming), and AI chatbot via Groq API
- **Appointment Organizer:** Centralized scheduling, document storage, and prescription management.
- **Pet Detection**
- **Technical Skills:** Java Spring boot, ReactJs, MongoDB, Python.

Energy Usage Prediction for Smart Grid Optimization

ML Project | [Project Link](#)

- **Energy Usage Prediction:** Predict short-term and long-term energy usage from time series data
- **Models:** Fine Tuned ARIMA, NBeats and GRU, LSTM, NNGA, NNPSO models to compare different metrics
- **Architecture:** LSTM model with NNPSO (Neural Network-based Particle Swarm Optimization) for better performance
- **Technical Skills:** Machine Learning, Python.

MediaSynergy

Software Project | [Front-End Code](#) | [Back-End Code](#)

- **Content Management:** Create, edit, save, and schedule posts for multiple platforms via Ayrshare API.
- **Analytics Dashboard:** Track post performance (comments, reach, reactions) using Facebook Graph API.
- **Business Community:** Dedicated space for startups and organizations to network and communicate.
- **Technical Skills:** ReactJs, NodeJs, MongoDB.

HomeSpace: A web app for seamless rental property booking

Database Project | [Code](#)

- **House entry & Tenants:** Homeowners can register and list their rental homes, and tenants can browse and book.
- **Chat System:** A messaging system facilitates personal communication between owners and tenants to finalize the deal.
- **Technical Skills:** NodeJs, JavaScript, Handlebars, CSS, Oracle

Skills

Languages	C/C++, Java, Python, JavaScript, HTML, CSS, SQL.
Frameworks & Database	NodeJs, ExpressJs, NextJs, Spring Boot, React, Postgresql
Developer & Documentation Tools	Git & Github, VS Code, Postman, $\LaTeX$, Google Colab, Kaggle.
Deep Learning	Pytorch, Numpy, Pandas, Matplotlib, Scikit-learn, OpenCV
Miscellaneous	Linux, Shell(Bash/Zsh), Cisco Packet Tracer

References available upon request.